

# Primal-Dual Optimization and Duality

DA5000W Linear Algebra and Optimization

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# Primal Optimization Problem

A general constrained optimization problem (*primal*):

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m, \\ & h_j(x) = 0, \quad j = 1, \dots, p. \end{aligned}$$

- $f(x)$ : objective function
- $g_i(x)$ : inequality constraints
- $h_j(x)$ : equality constraints

Introduce Lagrange multipliers:

- $\lambda_i \geq 0$  for inequality constraints
- $\mu_j \in \mathbb{R}$  for equality constraints

The **Lagrangian** is

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \mu_j h_j(x).$$

# Dual Optimization Problem

The **dual function** is defined as

$$q(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu).$$

It's always concave in  $(\lambda, \nu)$  and provides a lower bound on the primal optimal value

The **dual problem** is

$$\begin{aligned} \max_{\lambda, \mu} \quad & q(\lambda, \mu) \\ \text{s.t.} \quad & \lambda \geq 0. \end{aligned}$$

- Maximization of a concave function
- Often easier than the primal

# Duality as Upper Bounds

Consider the linear program:

$$\begin{aligned} \max_{x_1, x_2, x_3 \in \mathbb{R}} \quad & x_1 + x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 + 4x_3 \leq 3, \\ & x_1 + x_2 - 3x_3 \leq 1. \end{aligned}$$

Instead of solving the LP, we aim to prove lower and upper bounds on the optimal objective value.

This is primal LP of the form

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b, \end{aligned}$$

# Deriving the Dual from the Primal (Matrix Form)

Let  $u \in \mathbb{R}^m$  with  $u \geq 0$ .

Multiplying the constraints by  $u$  gives:

$$u^\top Ax \leq u^\top b.$$

If we require

$$A^\top u \geq c,$$

then

$$c^\top x \leq u^\top Ax \leq u^\top b.$$

Thus  $b^\top u$  is an upper bound on the primal objective for all feasible  $x$ .

To obtain the *best* upper bound, we minimize over all such  $u$ :

$$\begin{aligned} \min_{u \in \mathbb{R}^m} \quad & b^\top u \\ \text{s.t.} \quad & A^\top u \geq c \\ & u \geq 0. \end{aligned}$$

# Converting a primal to a dual problem

The dual is obtained by:

- Introduce one dual variable per constraint
- Minimize the weighted sum of right-hand sides
- Ensure the weighted constraints dominate the objective

Introduce dual variables:

$$u_1 \geq 0 \quad \text{for constraint 1}, \quad u_2 \geq 0 \quad \text{for constraint 2.}$$

We minimize the weighted sum of the right-hand sides:

$$\min 3u_1 + u_2.$$

This gives one condition per primal variable:

$$\begin{aligned} 2u_1 + u_2 &\geq 1 && \text{(coefficient of } x_1) \\ u_1 + u_2 &\geq 1 && \text{(coefficient of } x_2) \\ 4u_1 - 3u_2 &\geq 0 && \text{(coefficient of } x_3) \end{aligned}$$

# Dual Linear Program

Thus, the Dual LP can be formulated as,

$$\begin{array}{ll}\min_{u_1, u_2} & 3u_1 + u_2 \\ \text{s.t.} & 2u_1 + u_2 \geq 1, \\ & u_1 + u_2 \geq 1, \\ & 4u_1 - 3u_2 \geq 0, \\ & u_1, u_2 \geq 0.\end{array}$$

This is equivalent to writing,

$$\begin{array}{ll}\min_{u \geq 0} & b^\top u \\ \text{s.t.} & A^\top u \geq c.\end{array}$$

As the dual problem, using matrices for a general case.

# Primal and Dual Linear Programs

## Primal (P)

$$\begin{aligned} \max_{x \in \mathbb{R}^n} \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b \\ & x \in \mathbb{R}^n \end{aligned}$$

*Decision variables:*  $x$

*Feasible region:*  $Ax \leq b$

## Dual (D)

$$\begin{aligned} \min_{u \in \mathbb{R}^m} \quad & b^\top u \\ \text{s.t.} \quad & A^\top u \geq c \\ & u \geq 0 \end{aligned}$$

*Dual variables:*  $u$



# Primal–Dual Correspondence Table

| Maximization problem   | Minimization problem   |
|------------------------|------------------------|
| $\leq$ constraint      | variable $\geq 0$      |
| $=$ constraint         | unconstrained variable |
| $\geq$ constraint      | variable $\leq 0$      |
| variable $\geq 0$      | $\geq$ constraint      |
| unconstrained variable | $=$ constraint         |
| variable $\leq 0$      | $\leq$ constraint      |

Each constraint in the primal determines the sign of the corresponding dual variable, and each variable sign in the primal determines the type of constraint in the dual.

# Why Is the Dual Often Easier to Solve?

For a convex optimization problem, (primal - min, dual - max)

- Dual has fewer variables (2 instead of 3)
- Constraints are simpler to interpret
- Any feasible dual solution gives an immediate upper bound
- No need to find primal feasibility
- Primal feasibility  $\Rightarrow$  lower bounds
- Dual feasibility  $\Rightarrow$  upper bounds
- Dual = best way to combine constraints
- Solving the dual can certify optimality without solving the primal

# Primal–Dual Linear Program

## Primal (P):

$$\begin{array}{ll}\max_{x,y \geq 0} & 2x + 3y \\ \text{s.t.} & -x + y \leq 3 \\ & x - 2y \leq 2 \\ & x + y \leq 7\end{array}$$

We solve the primal using the **simplex method**.

Introduce slack variables  $s_1, s_2, s_3 \geq 0$ :

$$\begin{array}{rcl}-x + y + s_1 & = & 3 \\ x - 2y + s_2 & = & 2 \\ x + y + s_3 & = & 7\end{array}$$

Objective:

$$\max z = 2x + 3y$$

# Initial Simplex Tableau

Initial basic feasible solution:

$$x = y = 0, \quad s_1 = 3, \quad s_2 = 2, \quad s_3 = 7$$

|       | $x$ | $y$ | $s_1$ | $s_2$ | $s_3$ | RHS |
|-------|-----|-----|-------|-------|-------|-----|
| $s_1$ | -1  | 1   | 1     | 0     | 0     | 3   |
| $s_2$ | 1   | -2  | 0     | 1     | 0     | 2   |
| $s_3$ | 1   | 1   | 0     | 0     | 1     | 7   |
| $z$   | -2  | -3  | 0     | 0     | 0     | 0   |

Entering variable:  $y$  (most negative coefficient).

Minimum ratio test (positive entries in  $y$  column):

$$\frac{3}{1} = 3, \quad \frac{7}{1} = 7$$

Pivot element:  $(s_1, y)$ .

# Tableau After Pivot

|       | $x$ | $y$ | $s_1$ | $s_2$ | $s_3$ | RHS |
|-------|-----|-----|-------|-------|-------|-----|
| $y$   | -1  | 1   | 1     | 0     | 0     | 3   |
| $s_2$ | -1  | 0   | 2     | 1     | 0     | 8   |
| $s_3$ | 2   | 0   | -1    | 0     | 1     | 4   |
| $z$   | -5  | 0   | 3     | 0     | 0     | 9   |

Entering variable:  $x$ .

Minimum ratio test (positive entries in  $x$  column):

$$\frac{4}{2} = 2$$

Pivot element:  $(s_3, x)$ .

# Final Simplex Tableau

|       | $x$ | $y$ | $s_1$          | $s_2$ | $s_3$         | RHS |
|-------|-----|-----|----------------|-------|---------------|-----|
| $y$   | 0   | 1   | $\frac{1}{2}$  | 0     | $\frac{1}{2}$ | 5   |
| $s_2$ | 0   | 0   | $\frac{3}{2}$  | 1     | $\frac{1}{2}$ | 10  |
| $x$   | 1   | 0   | $-\frac{1}{2}$ | 0     | $\frac{1}{2}$ | 2   |
| $z$   | 0   | 0   | $\frac{1}{2}$  | 0     | $\frac{5}{2}$ | 19  |

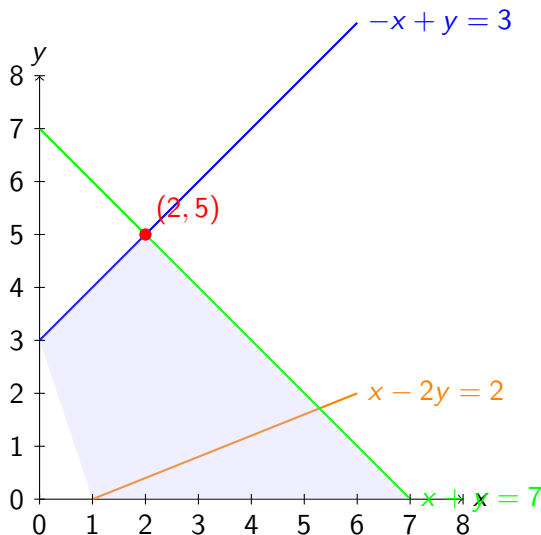
All reduced costs are nonnegative  $\rightarrow$  optimal.

The optimal solution is,

$$x^* = 2, \quad y^* = 5$$

$$z^* = 2x + 3y = 2(2) + 3(5) = 19$$

# Graphical Solution of the Primal LP



# KKT Conditions

For convex problems,  $x^*, \lambda^*, \nu^*$  are optimal iff:

- ① **Primal feasibility:**  $g_i(x^*) \leq 0, h_j(x^*) = 0$
- ② **Dual feasibility:**  $\lambda_i^* \geq 0$
- ③ **Complementary slackness:**

$$\lambda_i^* g_i(x^*) = 0$$

- ④ **Stationarity:**

$$\nabla f(x^*) + \sum_i \lambda_i^* \nabla g_i(x^*) + \sum_j \nu_j^* \nabla h_j(x^*) = 0$$



# Example: Simple Inequality Constraint

**Primal problem:**

$$\min_x x^2 \quad \text{s.t.} \quad x \geq 1.$$

Rewrite constraint as  $g(x) = 1 - x \leq 0$ .

Lagrangian:

$$\mathcal{L}(x, \lambda) = x^2 + \lambda(1 - x), \quad \lambda \geq 0.$$

Dual function:

$$q(\lambda) = \inf_x (x^2 + \lambda(1 - x)).$$

Minimizing over  $x$  gives  $x = \lambda/2$ .

$$q(\lambda) = \lambda - \frac{\lambda^2}{4}.$$

Dual problem:

$$\max_{\lambda \geq 0} \lambda - \frac{\lambda^2}{4}.$$

# Duality and Complementary Slackness

We are given the linear program:

$$\begin{array}{ll}\max & 3x_1 + 2x_3 \\ \text{s.t.} & x_1 + x_2 + x_3 \leq 6 \\ & 2x_1 - x_2 + x_3 \leq 3 \\ & 3x_1 + x_2 - x_3 \leq 3 \\ & x_1, x_2, x_3 \geq 0\end{array}$$

A candidate optimal solution is  $(x_1, x_2, x_3) = (0, 1.5, 4.5)$ .

We first verify feasibility:

- $x_1, x_2, x_3 \geq 0$  ✓
- $x_1 + x_2 + x_3 = 6$  ✓
- $2x_1 - x_2 + x_3 = 3$  ✓
- $3x_1 + x_2 - x_3 = -3 < 3$  ✓

Thus, the point is feasible and a valid candidate for optimality.

# Dual Problem

The dual of the primal LP is:

$$\begin{aligned} \min_{u_1, u_2, u_3 \geq 0} \quad & 6u_1 + 3u_2 + 3u_3 \\ \text{s.t.} \quad & u_1 + 2u_2 + 3u_3 \geq 3 \\ & u_1 - u_2 + u_3 \geq 0 \\ & u_1 + u_2 - u_3 \geq 2 \end{aligned}$$

At the primal point:

- Constraint 1:  $x_1 + x_2 + x_3 = 6$  (tight)
- Constraint 2:  $2x_1 - x_2 + x_3 = 3$  (tight)
- Constraint 3:  $3x_1 + x_2 - x_3 < 3$  (slack)

Since the last constraint is strictly slack (not tight), complementary slackness implies the corresponding dual variable must be zero:

$$u_3(3 - (3x_1 + x_2 - x_3)) = 0 \implies \boxed{u_3 = 0}.$$

# Complementary Slackness: Dual Constraints

Since the primal variables  $x_2 = 1.5 > 0$  and  $x_3 = 4.5 > 0$ , the corresponding dual constraints (D2) and (D3) must be binding (tight):

$$u_1 - u_2 + u_3 = 0,$$

$$u_1 + u_2 - u_3 = 2.$$

Substituting  $u_3 = 0$  into this system yields:

$$\begin{cases} u_1 - u_2 = 0 \\ u_1 + u_2 = 2 \end{cases}$$

Solving this simple system gives:

$$2u_1 = 2 \implies u_1 = 1, \quad \text{and} \quad u_2 = 1.$$

Thus, the candidate dual solution is:

$$(u_1, u_2, u_3) = (1, 1, 0).$$

# Dual Feasibility and Optimality

We must verify if  $(1, 1, 0)$  satisfies the remaining dual constraint :

$$u_1 + 2u_2 + 3u_3 = 1 + 2(1) + 3(0) = 3 \geq 3.$$

The constraint holds. Since  $(1, 1, 0)$  is dual feasible and satisfies complementary slackness with the feasible  $\bar{x}$ , both are optimal. The common optimal objective value is:

$$3(0) + 2(4.5) = 9.$$

# Characterization of All Primal Optimal Solutions

To find *all* optimal primal solutions, we note that the optimal dual variables are strictly positive ( $u_1 > 0$  and  $u_2 > 0$ ). By complementary slackness, this forces the first two primal constraints to be tight for *any* optimal solution:

$$\begin{aligned}x_1 + x_2 + x_3 &= 6, \\2x_1 - x_2 + x_3 &= 3.\end{aligned}$$

Let  $x_1 = t$  be the parameter. Subtracting the equations eliminates  $x_3$  temporarily, or we can solve generally. From equation 1:  $x_2 + x_3 = 6 - t$ . From equation 2:  $-x_2 + x_3 = 3 - 2t$ . Adding these two:

$$2x_3 = 9 - 3t \implies x_3 = 4.5 - 1.5t.$$

Substituting back to find  $x_2$ :

$$x_2 = (6 - t) - (4.5 - 1.5t) = 1.5 + 0.5t.$$

So the solutions are parametrized as:

$$(x_1, x_2, x_3) = (t, 1.5 + 0.5t, 4.5 - 1.5t).$$

# Feasibility Conditions

For this solution to be valid, it must satisfy non-negativity ( $x \geq 0$ ) and the third constraint (P3).

## ① Non-negativity:

$$t \geq 0$$

$$1.5 + 0.5t \geq 0 \implies t \geq -3$$

$$4.5 - 1.5t \geq 0 \implies 1.5t \leq 4.5 \implies t \leq 3$$

So currently  $0 \leq t \leq 3$ .

## ② Constraint (P3):

$$3x_1 + x_2 - x_3 \leq 3$$

$$3(t) + (1.5 + 0.5t) - (4.5 - 1.5t) \leq 3$$

$$3t + 1.5 + 0.5t - 4.5 + 1.5t \leq 3$$

$$5t - 3 \leq 3$$

$$5t \leq 6 \implies t \leq 1.2$$

Combining the conditions, the valid range for  $t$  is  $0 \leq t \leq 1.2$ .

- ❶ **Primal Solution** : The primal problem gave us the solution set:

$$\{(t, 1.5 + 0.5t, 4.5 - 1.5t) \mid 0 \leq t \leq 1.2\}.$$

Despite the multiple  $x^*$ , the maximum objective value is unique and constant at **9**.

- ❷ **Dual Solution**: The dual problem provided us with the solution:

$$(u_1^*, u_2^*, u_3^*) = (1, 1, 0).$$

The minimum dual cost is **9**, matching the primal solution.



# Role of Duality in Solving the Problem

Duality plays a fundamental role in solving the linear program by providing both a *global bound* on the objective value and a *certificate of optimality*. These are guarantees that the primal formulation alone does not directly provide.

By weak duality, for any primal feasible  $x$  and any dual feasible  $u$ , the following inequality holds:

$$3x_1 + 2x_3 \leq 6u_1 + 3u_2 + 3u_3.$$

Thus, every dual feasible point produces an *upper bound* on the primal objective. In particular, the dual feasible point

$$(u_1, u_2, u_3) = (1, 1, 0)$$

yields

$$3x_1 + 2x_3 \leq 6(1) + 3(1) + 3(0) = 9$$

for *all* primal feasible  $x$ .

# Conclusion

Therefore, duality proves that no feasible primal solution can have objective value exceeding 9.

Solving the primal problem directly would typically require enumerating vertices of the feasible polytope or executing the simplex algorithm.

Duality avoids this entirely. Specifically, duality:

- provides a *global optimality certificate* without vertex enumeration,
- explains *why* the objective cannot be increased further,
- identifies which constraints are binding through complementary slackness,
- characterizes *all* optimal solutions, not just one.